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(54) **OPTICAL FIBER CABLE TEMPERATURE MEASUREMENT DEVICE**

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(52) **U.S. Cl.**

CPC **G01K 1/143** (2013.01); **G01K 11/3206** (2013.01); **G01K 11/324** (2021.01); **H01M 10/486** (2013.01)

(58) **Field of Classification Search**

CPC .. G01K 1/143; G01K 11/324; G01K 11/3206; H01M 10/486

See application file for complete search history.

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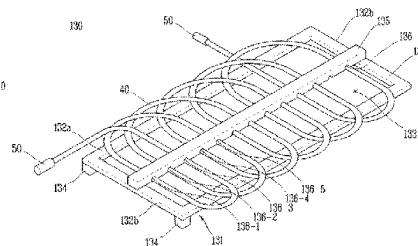
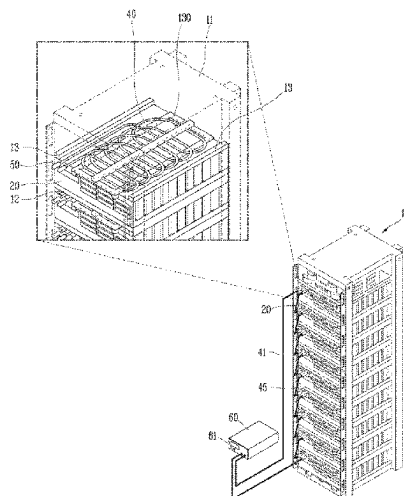
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(57) **ABSTRACT**

The present invention relates to an optical fiber cable temperature measurement device, and more particularly, an optical fiber cable temperature measurement device that is applied to a multi-stage power device module such as a battery storage device. An optical fiber cable temperature measurement device according to an embodiment of the present invention is applied to a power device including a plurality of power device modules, and comprises: optical fiber cable measurement units that are provided in the respective power device modules; optical fiber cables inserted and installed in the respective optical fiber cable measurement units; and a control unit that is connected to one end of the optical fiber cables to measure the temperature.

22 Claims, 15 Drawing Sheets



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H01M 10/48 (2006.01)

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FIG. 1

Prior Art

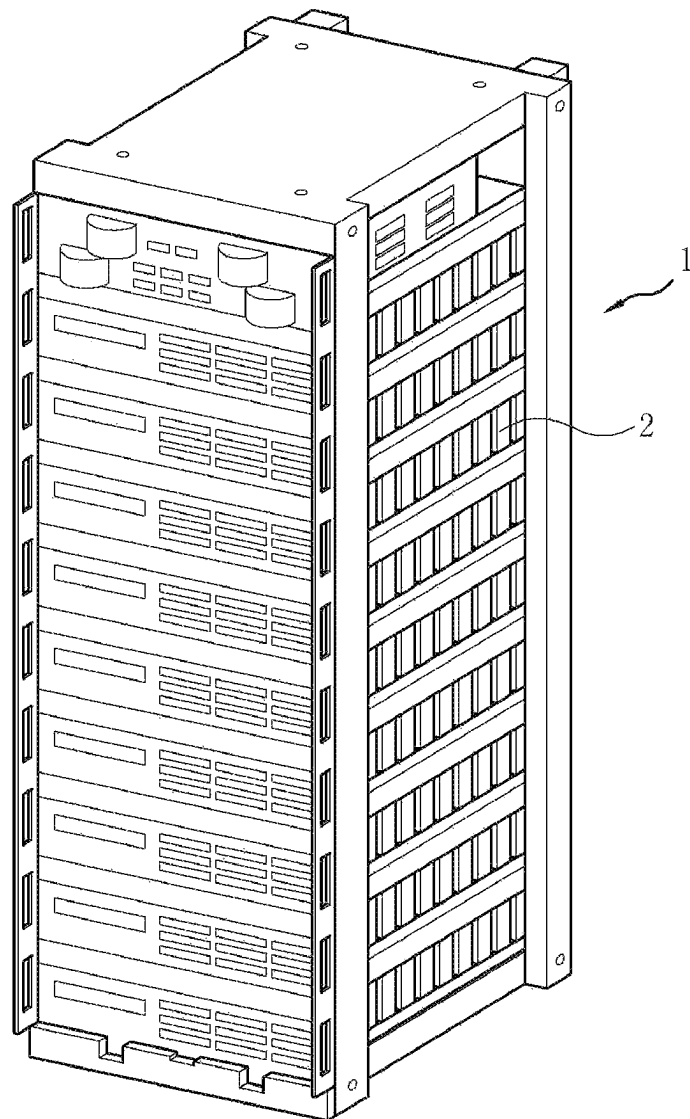


FIG. 2

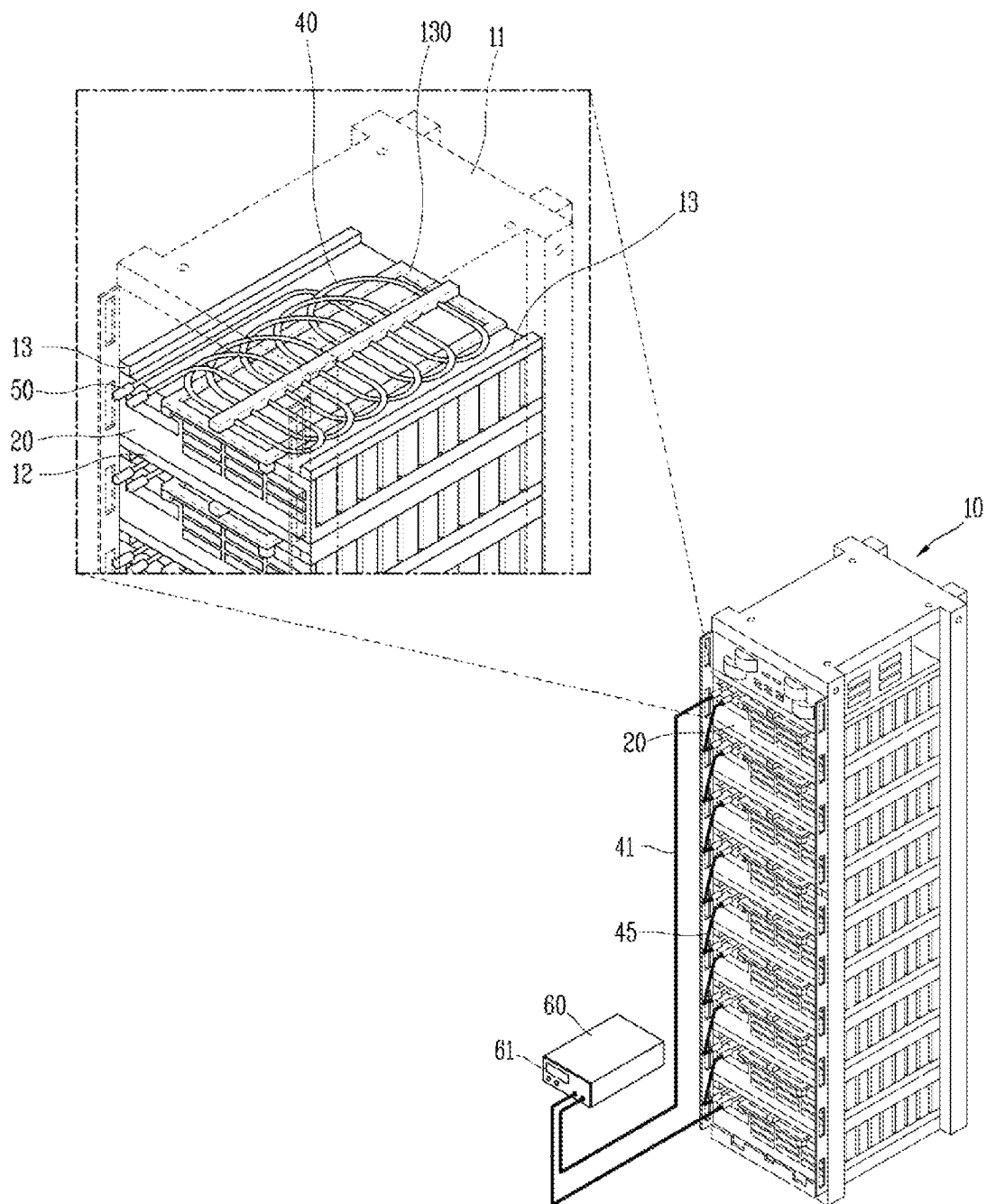


FIG. 3

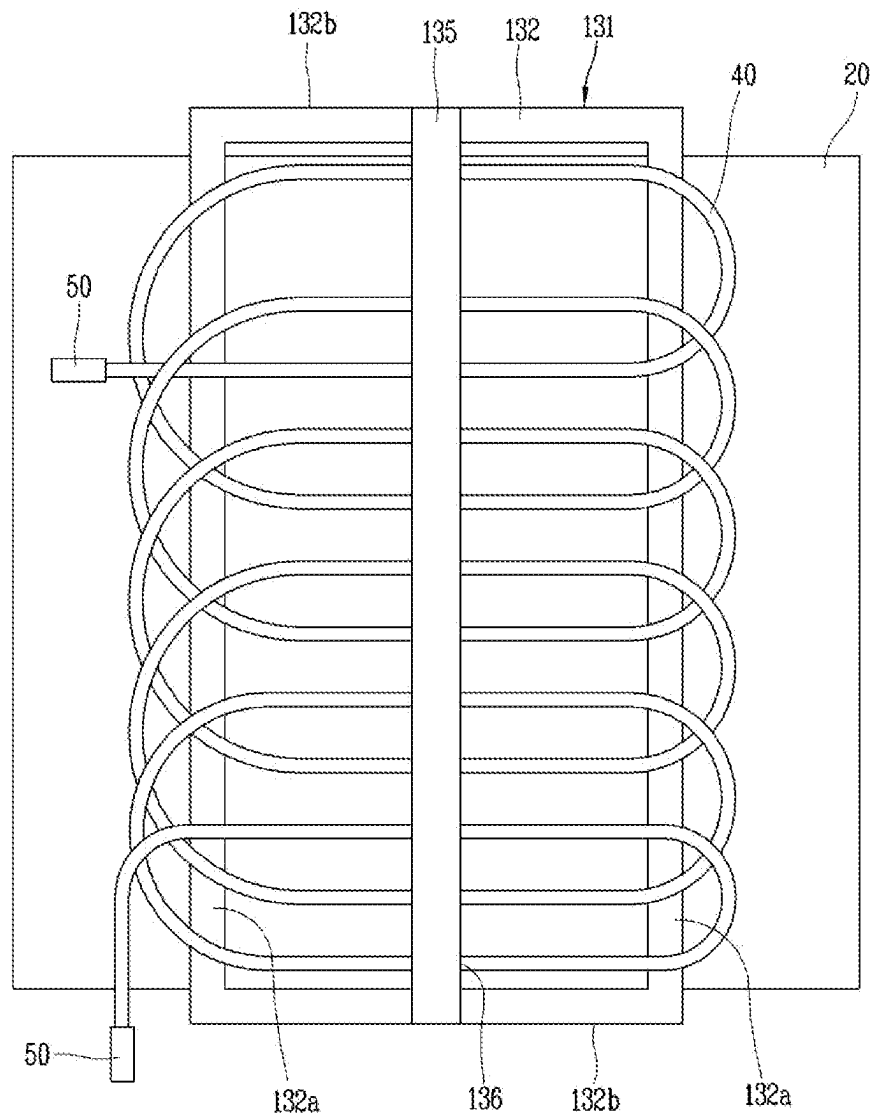


FIG. 4

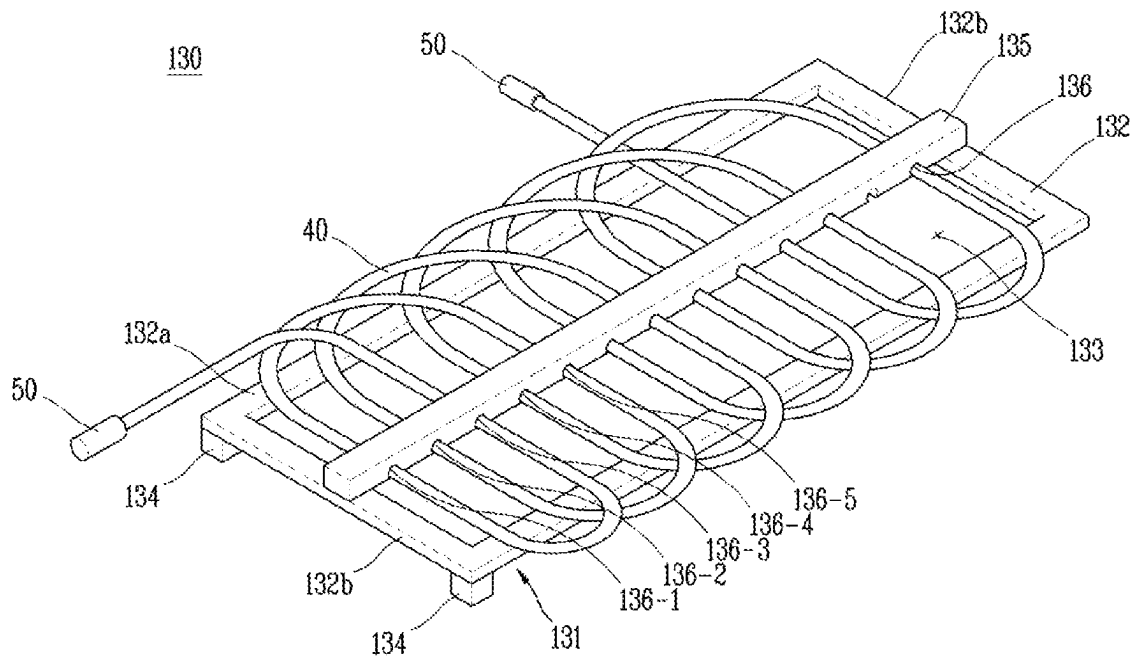


FIG. 5

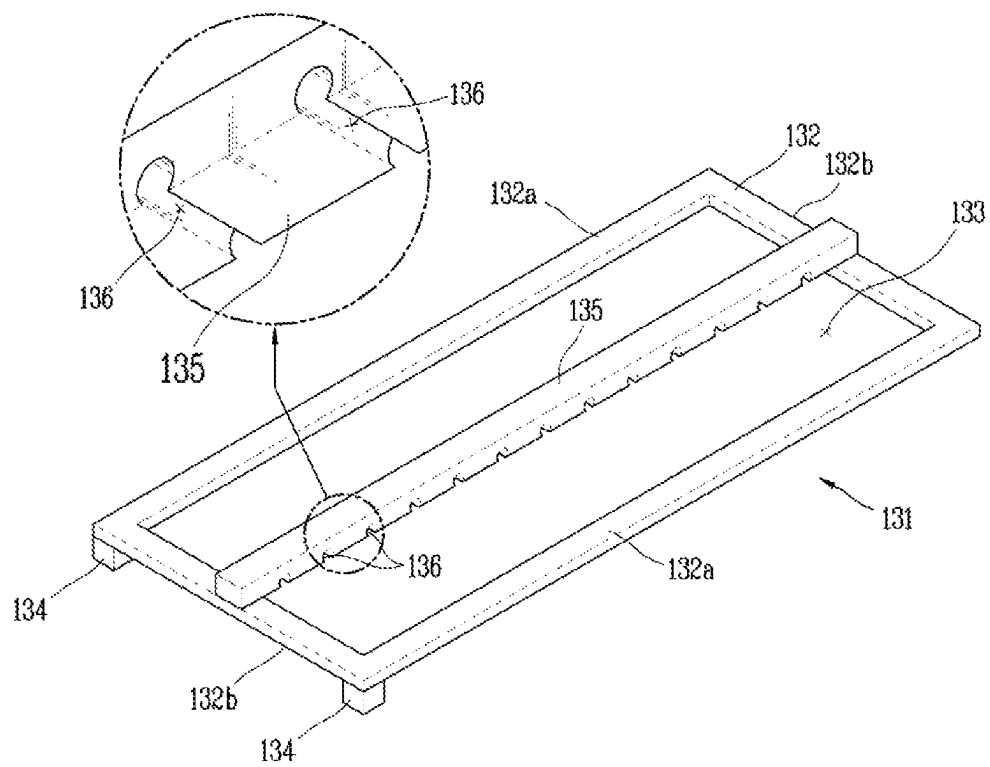


FIG. 6

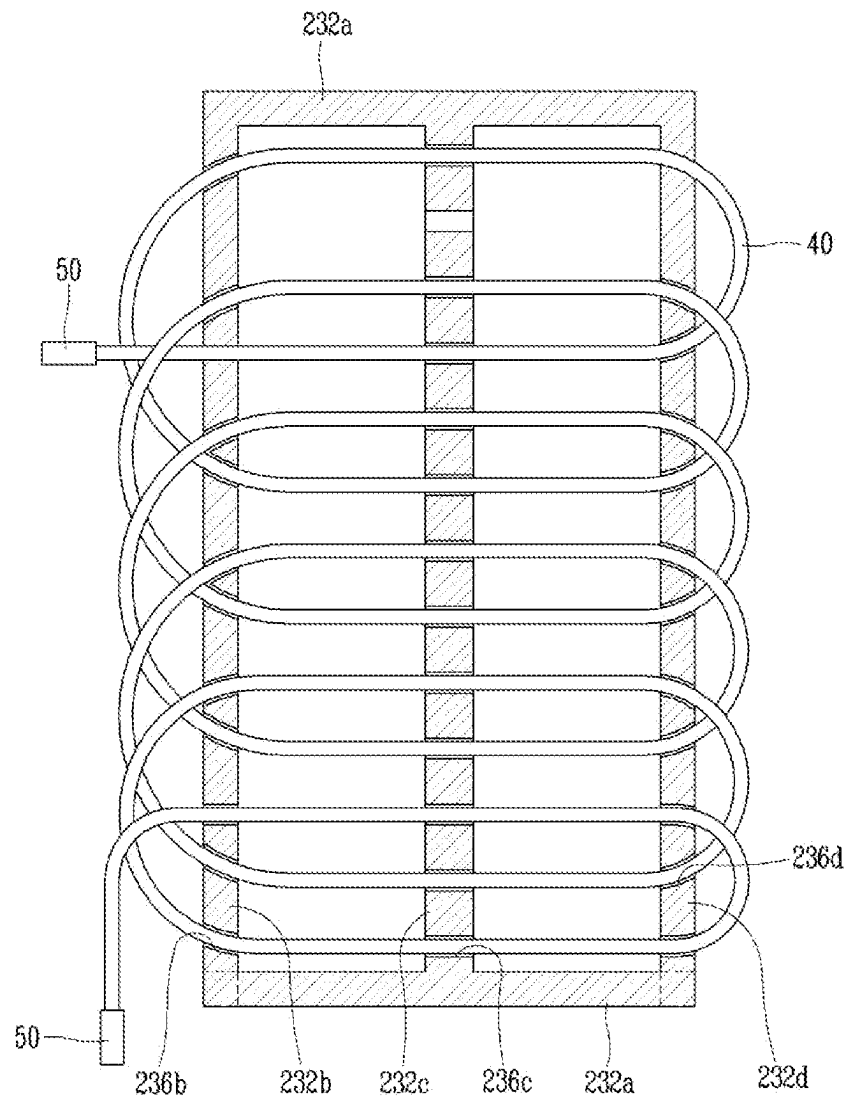


FIG. 7

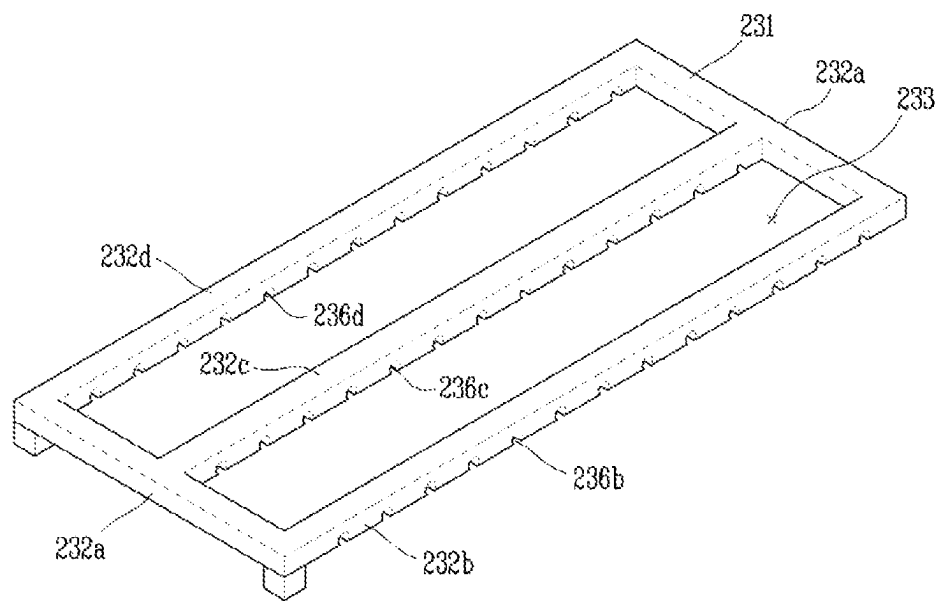


FIG. 8

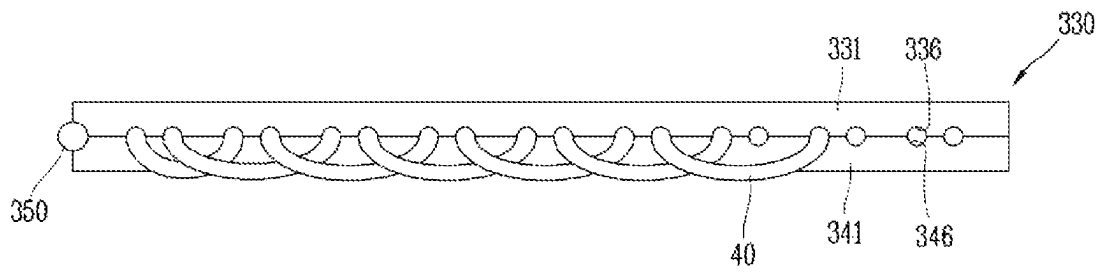


FIG. 9

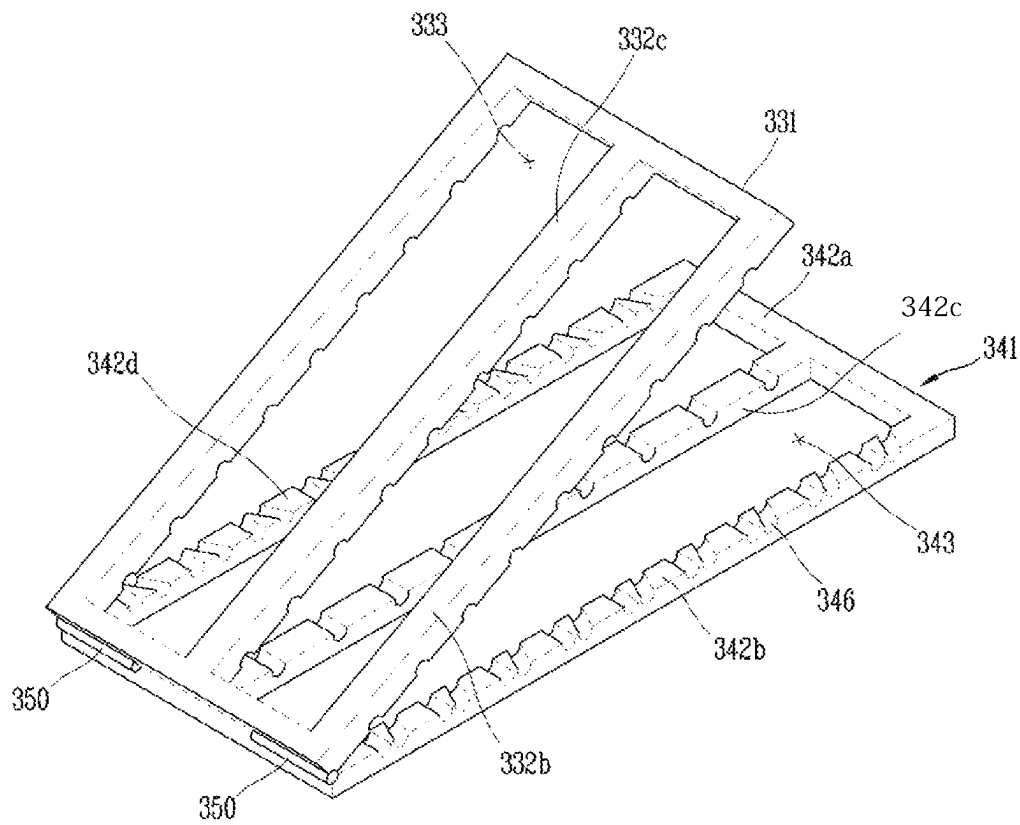


FIG. 10

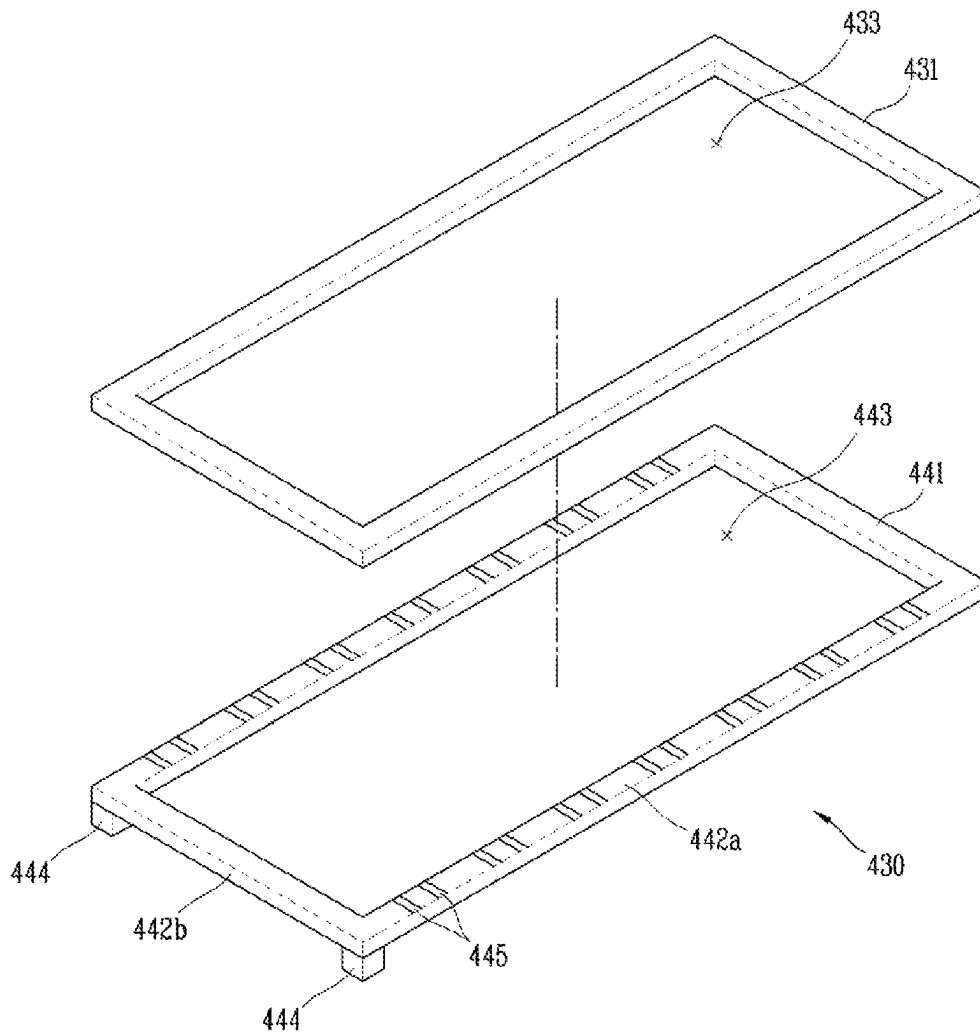


FIG. 11

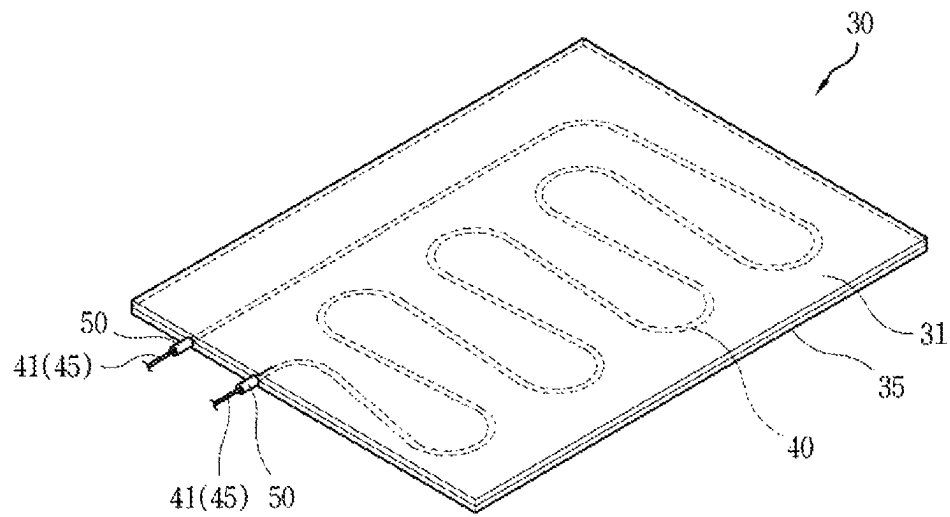


FIG. 12

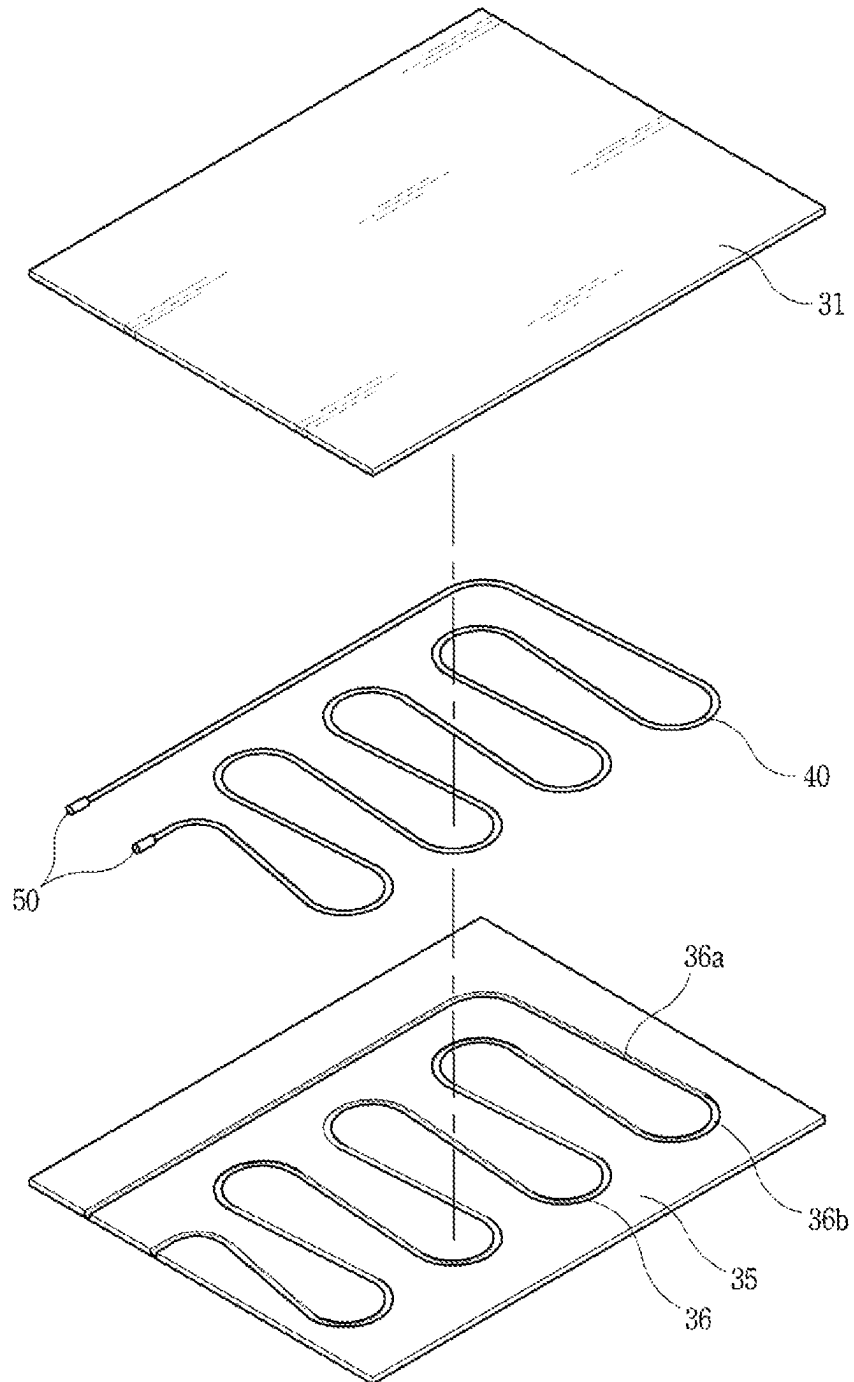


FIG. 13

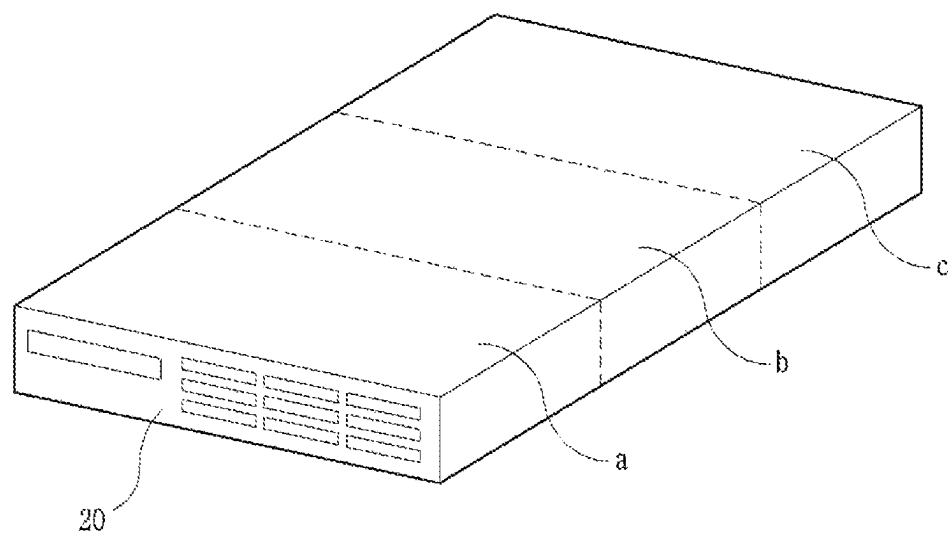


FIG. 14

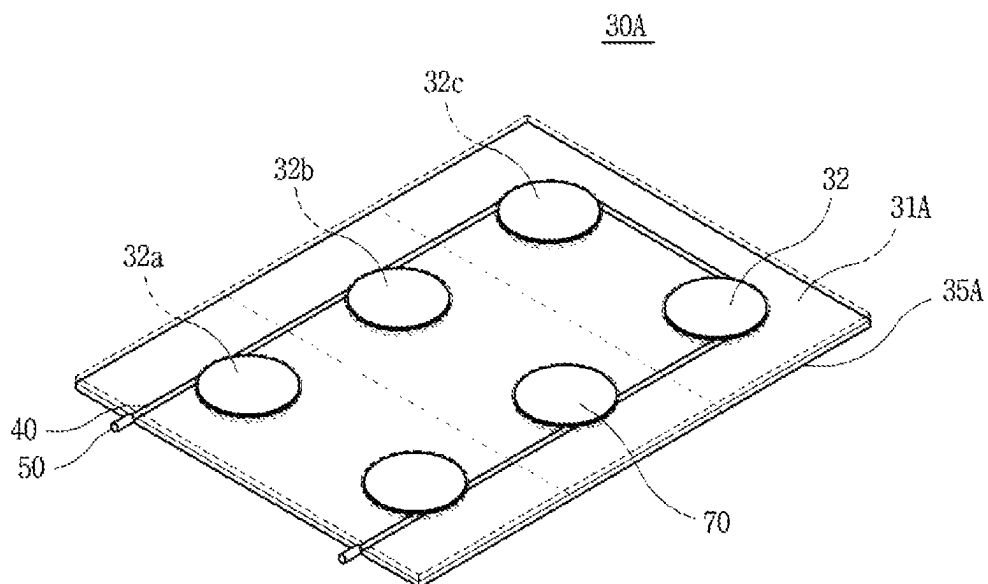


FIG. 15

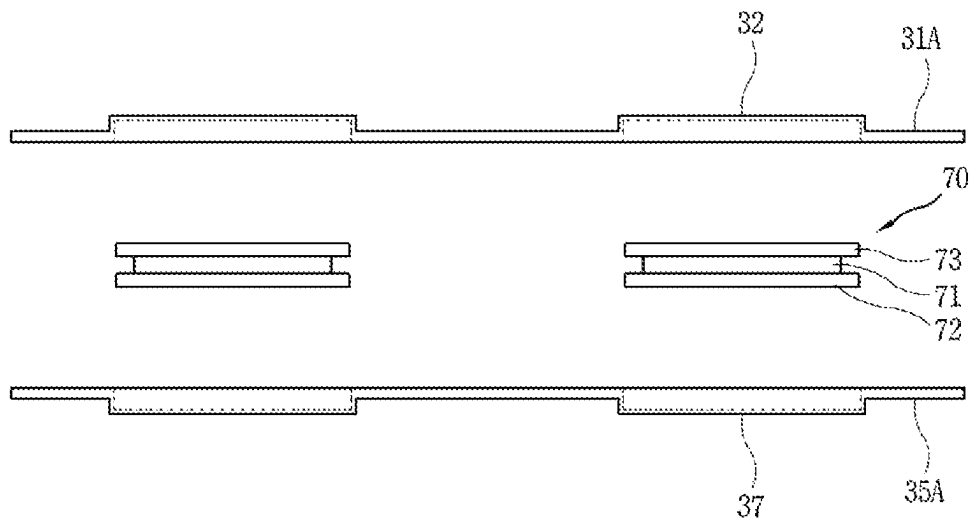


FIG. 16

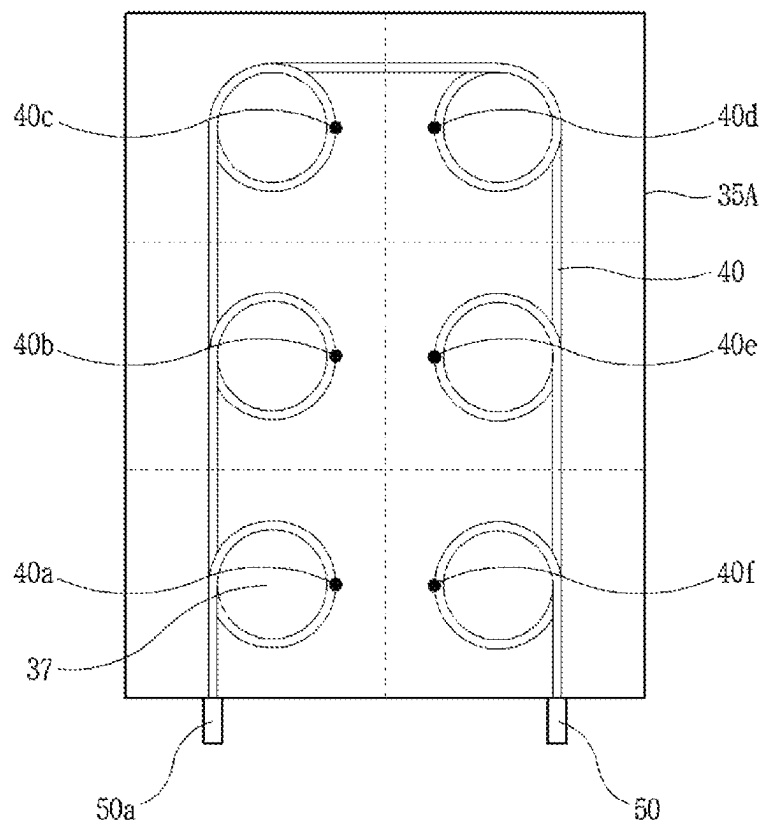
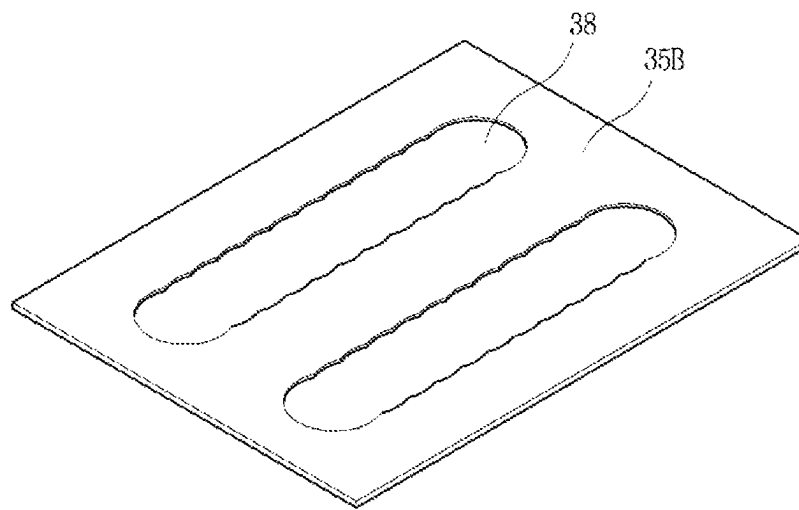


FIG. 17



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OPTICAL FIBER CABLE TEMPERATURE MEASUREMENT DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2020/015734, filed on Nov. 11, 2020, which claims the benefit of earlier filing date and right of priority to Korean Application Nos. 10-2019-0161001, filed on Dec. 5, 2019, 10-2020-0063237, filed on May 26, 2020, the contents of which are all hereby incorporated by reference herein in their entirety.

FIELD

The present disclosure relates to an optical fiber cable temperature measurement device, and more particularly, to an optical fiber cable temperature measurement device that is applied to a multi-stage power device module such as a battery storage device.

BACKGROUND

Generally, an energy storage system (ESS) refers to a device that stores energy using a physical medium. Energy storage is largely divided into physical energy storage and chemical energy storage according to a storage method. Typical physical energy storage includes pumping-up power generation, compressed air storage, and a flywheel etc., and the chemical energy storage mainly uses batteries, such as lithium-ion batteries, lead acid batteries, and sodium-sulfur (NaS) batteries.

As an example of an energy storage device, battery type ESS is called BESS (Battery Energy Storage System), and in general, ESS means BESS.

Energy storage devices using batteries generate much heat, so it is important to manage heat to prevent fires. To this end, the energy storage device is usually provided with a temperature sensor therein.

Referring to FIG. 1, an energy storage device (ESS) according to the related art will be described. An energy storage device using batteries usually comprises cells, modules, or racks. Certification of energy storage devices is done on a per-rack basis. The energy storage device using batteries is built in the form of a rack 1. The rack 1 is composed of battery modules 2 stacked in multiple stages on a cabinet-type structure made of beams or iron plates. Here, each battery module 2 is a combination of multiple battery cells (not shown). Each battery module 2 usually has a temperature sensor (not shown) inside for measuring their own temperatures.

However, when the temperature sensor breaks down, the temperature measurement of the battery module 2 is not possible. Also, measuring temperatures in the space between adjacent battery modules 2 is not possible either since the temperature sensor is provided inside the battery module 2. That is, temperature management for the rack 1 is not done because temperature measurement around the battery module 2 is not done.

SUMMARY

The present disclosure has been devised to solve the above problems, and an object of the present disclosure is to

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provide an optical fiber cable temperature measurement device capable of measuring temperatures around a multi-stage power device module.

Another object of the present disclosure is to provide an optical fiber cable temperature measurement device capable of measuring each of power device modules arranged in multiple stages.

Another object of the present disclosure is to provide an optical fiber cable temperature measurement device that is easy to assemble.

Another object of the present disclosure is to manage the heat of an energy storage device by measuring temperatures around a battery module of the multi-stage energy storage device.

To achieve these and other advantages and in accordance with the purpose of the present invention, as embodied and broadly described herein, there is provided an optical fiber cable temperature measurement device comprising: a power device provided with a plurality of power device modules; a plurality of optical fiber cable measurement units that are provided in the respective power device modules; optical fiber cables inserted and installed in the respective optical fiber cable measurement units; and a control unit that is connected to one end of the optical fiber cables to measure temperatures.

The optical fiber cable measurement units each comprise a cable mount where the optical fiber cable is inserted and mounted.

The cable mount comprises a mount body built in the shape of a rectangular frame and a crossbar disposed across the mount body.

The cable mount has a plurality of openings.

The mount body comprises a pair of horizontal bars disposed in parallel and a pair of vertical bars disposed in parallel.

The crossbar is disposed in the middle part of the pair of horizontal bars.

The mount body has a stopper that keeps the cable mount from entering when inserted into the power device module.

A plurality of cable fixing grooves are formed on a lower surface of the crossbar, for fixing the optical fiber cable.

A side cross-section of the cable fixing grooves is shaped like a segment of a circle.

The cable fixing grooves are formed on the vertical bars.

The cable fixing grooves formed on the crossbar are parallel to the horizontal bars, and the cable fixing grooves formed on the vertical bars are slanted from the horizontal bars.

The cable mount comprises an upper cable mount built in the form of a frame and a lower cable mount built in the form of a frame, corresponding to the upper cable mount.

The upper cable mount and the lower cable mount are held together in such a way as to swing relative to each other.

Positional marks where the optical fiber cable is mounted are formed on the lower cable mount or the upper cable mount.

The optical fiber cable temperature measurement device comprises an upper panel member and a lower panel member, and the optical fiber cable is inserted and installed between the upper panel member and the lower panel member.

The optical fiber cable temperature measurement device further comprises a connector coupled to both ends of the optical fiber cable and exposed out of the upper panel member and the lower panel member.

The upper panel member and the lower panel member are made of a thin plate or a film.

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The upper panel member or the lower panel member has an insertion groove formed therein where the optical fiber cable is inserted and installed, wherein the insertion groove has a straight portion or a curved portion.

The radius of curvature of the curved portion is 20 times as large as or larger than the diameter of the optical fiber cable.

The optical fiber cable temperature measurement device further comprises a connecting cable for connecting the control unit and the connector or connecting the connectors of adjacent battery modules.

The optical fiber cable temperature measurement device further comprises mounting plate installation parts that are formed as grooves on the upper panel member and the lower panel member and inserted into the mounting plate installation parts.

The mounting plate has a wiring groove formed around the circumference in such a way the optical fiber cable is wound thereabout.

The battery module is divided into a plurality of sections, and measurement points are set on the optical fiber cable at predetermined distances from the connector.

According to an optical fiber cable temperature measurement device according to an embodiment of the present disclosure, it is possible to facilitate temperature measurement for a power device and thermal control by an optical fiber cable measurement unit provided for each power device module.

Moreover, the optical fiber cable temperature measurement device may be applied for each power device module configured in multiple stages, by means of optical fiber cable measurement units.

Furthermore, the optical fiber cable temperature measurement device is provided in the form of optical fiber cable measurement units where the fiber-optic cable is mounted, and therefore may be easily assembled and installed in a power device.

Furthermore, the optical fiber cable measurement units make measurements on a per-sheet or per-plane basis, thereby enabling effective control of temperatures between battery modules.

Furthermore, the optical fiber cable measurement units allow for measurement using a fiber-optic cable that is inserted and laid between films (upper and lower panels) that are tightly held together, and therefore provides high measurement accuracy and enables conversion into measurement positions based on distance, thereby facilitating the arrangement and control of the measurement points.

Since the fiber-optic cable is laid on a film surface with a predetermined area, it is possible to secure a radius of curvature required for measurement.

Since the fiber-optic cable is installed in an installation groove or a mounting plate formed (placed) on the film surface, a standardized temperature measurement position distribution can be configured, thereby allowing for efficient temperature management.

Since the upper and lower panels may work in one direction by using an insulating/non-insulating film, the directionality of temperature measurement can be achieved.

Since temperature measurement is possible with respect to a connector provided for each optical fiber cable measurement unit. Thus, measurements may be made at the same points on the upper surfaces of battery modules even if the gap between the battery modules varies.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a power device according to the related art.

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FIG. 2 is a perspective view of a power device with an optical fiber cable temperature measurement device according to an embodiment of the present disclosure.

FIG. 3 is a plan view of the optical fiber cable temperature measurement device being installed on any one of power device modules in FIG. 2.

FIG. 4 is a perspective view of an optical fiber cable measurement unit according to an embodiment of the present disclosure.

FIG. 5 is a perspective view of the optical fiber cable measurement unit in FIG. 4.

FIG. 6 is a bottom view of an optical fiber cable temperature measurement device according to another embodiment of the present disclosure.

FIG. 7 is a perspective view of the optical fiber cable measurement unit in FIG. 6.

FIG. 8 is a side view of an optical fiber cable measurement unit according to yet another embodiment of the present disclosure.

FIG. 9 is a perspective view of the optical fiber cable measurement unit in FIG. 8.

FIG. 10 is a perspective view of an optical fiber cable measurement unit according to a further embodiment of the present disclosure.

FIGS. 11 and 12 are a perspective view and an exploded perspective view of an optical fiber cable measurement unit according to a further embodiment of the present disclosure.

FIG. 13 is a perspective view of a battery module in FIG. 2.

FIGS. 14 and 15 are a perspective view and an exploded perspective view of an optical fiber cable measurement unit according to a further embodiment of the present disclosure.

FIG. 16 is a top view of the optical fiber cable measurement unit in FIG. 14 from which the upper panel member and the mounting plate are removed.

FIG. 17 is a perspective view of a lower panel applied to an optical fiber cable temperature measurement device according to a further embodiment of the present disclosure.

DETAILED DESCRIPTION

Hereinafter, preferred embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. This is intended to describe the present disclosure in detail so that those having ordinary knowledge in the art to which the present disclosure pertains can easily practice the present disclosure, but does not mean that the technical spirit and scope of the present disclosure are limited by the above embodiments.

An optical fiber cable temperature measurement device according to each of the embodiments of the present disclosure will be described in detail with reference to the drawings.

First Embodiment

An optical fiber cable temperature measurement device according to an embodiment of the present disclosure includes: a power device 10 provided with a plurality of power device modules 20; a plurality of optical fiber cable measurement units 130 that are provided in the respective power device modules 20; optical fiber cables 40 inserted and installed in the respective optical fiber cable measurement units 130; and a control unit 60 that is connected to one end of the optical fiber cables 40 to measure temperatures.

The power device 10 is provided. The power device 10 may be configured in the form of a rack 11 so that a plurality

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of power device modules **20** can be installed in multiple stages. The rack **11** may include an upper plate and a lower plate and a frame connected between the upper plate and the lower plate to support them. A plurality of power device modules **20** are installed in multiple stages in the rack **11**. Here, an example of the power device may include an energy storage device. In addition, an example of the power device modules may include battery modules.

The power device **10** is shown in FIG. 2. The power device **10** is configured in the form of a rack as described above.

A plurality of support plates **12** are provided in the rack **11** so that each power device module **20** is installed and supported on them.

In addition, the rack **11** is provided with a support plate **13** so that the optical fiber cable measurement units **130** are installed on it. The support plate **13** may be provided above or below each power device module **20**. Accordingly, the optical fiber cable measurement units **130** may be inserted and installed between the power device modules **20** and the support plate **13**.

An optical fiber cable measurement unit **130** is shown in FIGS. 3 and 4. Also, a cable mount **131** is shown in FIG. 5. The optical fiber cable measurement unit **130** is provided to measure the temperature in the space between the power device module **20** and an adjacent power device module **20**. The optical fiber cable measurement unit **130** is disposed in such a way as to have surface contact with one surface (e.g., upper surface) of the power device module **20** or to have a predetermined gap with it, to measure the surface temperature of the power device module **20** or the temperature in the space between power device modules **20**.

The optical fiber cable measurement unit **130** may include a cable mount **131**, a fiber-optic cable **40**, and a connector **50**.

The cable mount **131** is provided. The cable mount **131** includes a mount body **132** built in the shape of a frame and a crossbar **135** disposed across the mount body.

The mount body **132** may be made of synthetic resin or metal. In this case, the mount body **132** is preferably made of a material having high thermal resistance and high thermal conductivity. Accordingly, the mount body **132** helps to dissipate heat from the power device module **20** while in contact with the upper surface of the power device module **20**.

The mount body **132** is built in the form of a polygonal (e.g., rectangular) frame. That is, the mount body **132** is mostly an opening **133**, except for the outer part. The opening **133** reduces the area of the power device module **20** covered by the mount body **132** and provides good air ventilation.

The mount body **132** may be part of a plain plate. Alternatively, the mount body **132** may be formed by bending a bar or a long plain plate.

The mount body **132** provides good air ventilation because of the opening **133**. When viewed from the top, the opening **133** covers most of the area of the mount body **132**, and therefore the thermal resistance the power device module **20** gets from the mount body **132** is minimized.

The mount body **132**, if built as a rectangular frame, includes a pair of horizontal bars **132b** disposed in parallel and a pair of vertical bars **132a** disposed in parallel.

The horizontal bars **132b** are preferably smaller than the width, i.e., horizontal length, of the power device module **20**.

The vertical bars **132a** may be larger than the depth, i.e., vertical length, of the power device module **20**.

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The mount body **132** has a stopper **134** that keeps the cable mount **131** from entering when inserted into the power device module **20**. The stopper **134** may protrude downward from both ends of the lower horizontal bar **132b**.

The mount body **132** is provided with a crossbar **135**. The crossbar **135** may be disposed between the upper horizontal bar **132b** and the lower horizontal bar **132b**. For example, the crossbar **135** may be disposed between the middle of the upper horizontal bar **132b** and the middle of the lower horizontal bar **132b**.

A plurality of cable fixing grooves **136** are formed on a lower surface of the crossbar **135**. The cable fixing grooves **136** are formed widthwise on the lower surface of the crossbar **135**.

A side cross-section of the cable fixing grooves **136** is shaped like a segment of a circle. Inlets of the cable fixing grooves **136** may be smaller than the diameter of the fiber-optic cable **40**. This keeps the fiber-optic cable **40** from falling out of the cable fixing grooves **136** after being force-fitted into the cable fixing grooves **136**.

The fiber-optic cable **40** is provided. The fiber-optic cable **40** is inserted and installed between the mount body **132** and the crossbar **135**. The fiber-optic cable **40** is laid over the mount body **132**. The fiber-optic cable **40** is laid in such a way as to be wound or bent between the mount body **132** and the crossbar **135**. That is, the fiber-optic cable **40** is laid in such a way that one cable is continuously wound. The fiber-optic cable **40** may overlap when bent.

As such, the temperature around the power device module **20** may be measured through the fiber-optic cable **40** laid on the cable mount **131**. As the fiber-optic cable **40** is a line, it may have a plurality of measurement points when laid on the surface of the cable mount **131**.

Temperature measurement using fiber-optic cables is conventionally employed in temperature measurement in conventional power underground lines and oil pipelines for refined oil and petrochemicals in the conventional art, which is also called Distributed Temperature Sensing (DTS).

Distributed temperature sensing uses the proportional relationship between temperature and wavelength, which is one of the unique characteristics of optical fibers.

There are three types of scatterings waves reflected from quartz from which optical fibers are made: Rayleigh-scattering waves, Raman-scattering waves, and Brillouin-scattering waves. Among them, the Raman waves represent a wavelength that is directly proportional to temperature. Using this, temperatures can be measured (measured wavelengths can be converted into temperatures depending on the amplitude). Since optic-fiber cables are linear, temperature measurement can be done by measuring the distance of each scattering frequency from a reference position (or by measuring each scattering frequency for each distance).

Temperature measurement using fiber-optic cables can be done at a resolution of 0.01° C. per 1 m. Also, temperature measurement can be done at intervals of 0.5 m or more for meaningful analysis of scattering waves.

Accordingly, the fiber-optic cable **40** is laid in a bent shape to ensure that a plurality of measurement points are provided within the limited area of an upper portion of the power device module **20**.

The fiber-optic cable **40** is laid on the vertical bars **132a** of the mount body **132** and fixed to the cable fixing grooves **136** of the crossbar **135**.

An example of a method of installing the fiber-optic cable **40** on the mount body **132** will be described below. As the fiber-optic cable **40** is wound either clockwise or counter-clockwise, it moves forward in one direction of the crossbar

135. The fiber-optic cable **40** may be disposed in such a way that it is wound multiple times in the form of a spring and then collapsed to one side. The fiber-optic cable **40** may be wound in an elliptical shape.

The fiber-optic cable **40** is laid on a pair of vertical bars **132a**. That is, the horizontal width (diameter) across which the fiber-optic cable **40** is wound may be longer than the horizontal bars **132b**.

The fiber-optic cable **40** may be disposed in such a way as to overlap each other as it is wound. That is, referring to FIGS. **3** to **5**, the fiber-optic cable **40** may be fitted into the cable fixing grooves **136** in an alternating manner. Suppose that the cable fixing grooves **136** are referred to as a first cable fixing groove **136-1**, a second cable fixing groove **136-2**, . . . , and an n-th cable fixing groove **136-n**, in the order of closest to the lower horizontal bar **132b**. Then, the fiber-optic cable **40** fixed to the cable fixing grooves **136** may be fitted in the order of the third cable fixing groove **136-3**, the first cable fixing groove **136-1**, the fifth cable fixing groove **136-5**, the second cable fixing groove **136-2**, the seventh cable fixing groove **136-7**, the fourth cable fixing groove **136-4**, the ninth cable fixing groove **136-9**, the sixth cable fixing groove **136-6**, From a second turn and onward of the fiber-optic cable **40**, the rule of alternation between moving 5 grooves forward and moving 3 grooves backward applies toward the upper end of the crossbar **135**.

The fiber-optic cable **40** is a cable that is installed on the cable mount **131** as it is wound multiple times.

The connector **50** is connected to the front and rear ends of the fiber-optic cable **40**. Thus, one optical fiber cable measurement unit **130** is completed.

The connector **50** is provided on both ends of the fiber-optic cable **40**. As the connector **50** is provided on each panel member **31** and **35**, the optical fiber cable measurement units **130** may be provided as separate units. Also, the connector **50** may be set as a reference point for temperature measurement (wavelength measurement). Accordingly, it may be set as a reference point of each optical fiber cable measurement unit **130**.

A connecting cable **41** and **45** is provided to connect the control unit **60** and the connector **50** or two adjacent connectors **50**. The optical fiber cable measurement units **130** may be interconnected by the connecting cable **45**.

The control unit **60** may be configured as a DTS server. The control unit **60** is provided to provide wavelength measurements in the fiber-optic cable **40** and convert them into temperatures.

A proper operating temperature may be set on the control unit **60**, and an indication or an alert may be issued if this temperature is out of a limited temperature range. For example, a display unit **61** may be provided on a portion of the server of the control unit **60**.

Second Embodiment

FIGS. **6** and **7** illustrate an optical fiber cable temperature measurement device according to another embodiment of the present invention. FIG. **6** is a bottom view of an optical fiber cable measurement unit according to a second embodiment, and FIG. **7** is a perspective view of a cable mount.

The optical fiber cable temperature measurement device includes: a power device **10** provided with a plurality of power device modules **20**; a plurality of optical fiber cable measurement units **230** that are provided in the respective power device modules **20**; optical fiber cables **40** inserted and installed in the respective optical fiber cable measure-

ment units **230**; and a control unit **60** that is connected to one end of the optical fiber cables **40** to measure temperatures.

In this embodiment, differences from the previous embodiment will be described.

The optical fiber cable measurement unit **230** includes a cable mount **231** and a fiber-optic cable **40**. The cable mount **231** is built in the shape of a frame with an opening **233**. For example, FIGS. **6** and **7** illustrate a cable mount **231** built in the form of a rectangular frame.

The cable mount **231** have a plurality of openings **233**. For example, the cable mount **231** includes two horizontal bars **232a** and three vertical bars **232b**, **232c**, and **232d**, with two openings **233** formed therein.

The vertical bars **232b**, **232c**, and **232d** may be divided into side vertical bars **232b** and a center vertical bar **232c**. Cable fixing grooves **236b**, **236c**, and **236d** are respectively formed on the lower surfaces of the vertical bars **232b**, **232c**, and **232d**.

The fiber-optic cable **40** is fixed by being fitted to the cable fixing grooves **236b**, **236c**, and **236d**. The fiber-optic cable **40** may be wound in such a way that it is fitted sequentially to the cable fixing groove **236b** of one side vertical bar **232b**, the cable fixing groove **236c** of the center vertical bar **232c** and then moved up and fitted sequentially to the cable fixing groove **236d** of the other side vertical bar **232d**, the cable fixing groove **236c** of the center vertical bar **232c**, and the cable fixing groove **236b** of the one vertical bar **232b**.

Here, the cable fixing grooves **236c** of the center vertical bar **232c** are parallel to the horizontal bars, and the cable fixing grooves **236b** and **236d** of the side vertical bars **232b** and **232d** are slanted from the horizontal bars.

The cable fixing grooves **236b** and **236d** of the side vertical bars **232b** and **232d** may be arranged in consecutive pairs of cable fixing grooves facing each other in a wedge-like pattern. Here, the cable fixing grooves **236b** of the one side vertical bar **232b** and the cable fixing grooves **236d** of the other side vertical bar **232d** may be formed at different elevations.

In this embodiment, the cable fixing grooves are formed on each of the vertical bars, thereby fixing and holding the fiber-optic cable **40** steady in place.

Third Embodiment

FIGS. **8** and **9** illustrate an optical fiber cable temperature measurement device according to yet another embodiment of the present disclosure.

The optical fiber cable temperature measurement device includes: a power device **10** provided with a plurality of power device modules **20**; a plurality of optical fiber cable measurement units **330** that are provided in the respective power device modules **20**; optical fiber cables **40** inserted and installed in the respective optical fiber cable measurement units **330**; and a control unit **60** that is connected to one end of the optical fiber cables **40** to measure temperatures.

In this embodiment, differences from the previous embodiment will be described.

The optical fiber cable measurement unit **330** includes a cable mount **331** and **341** and a fiber-optic cable **40**. The cable mount **331** and **341** includes an upper cable mount **331** and a lower cable mount **341** which are held together in such a way as to be opened or closed. For example, the upper cable mount **331** and the lower cable mount **341** are held together by a pivotal connecting member **350** such as a hinge about which they swing relative to each other.

The upper cable mount **331** and the lower cable mount **341** are built in the form of a frame with an opening **333** and **343**. The upper cable mount **331** may be built in the form described in the second embodiment. The lower cable mount **341** may be built in the form corresponding to the upper cable mount **331**. That is, they may be built in the form of a rectangular frame with a plurality of horizontal bars **342a** and a plurality of vertical bars **342b**, **342c**, and **342d**.

Cable mounting grooves **346** corresponding to cable mounting grooves **336** of the upper cable mount **331** are formed on the lower cable mount **341**.

The fiber-optic cable **40** is inserted and installed between the cable mounting grooves of the upper cable mount **331** and the cable mounting grooves of the lower cable mount **341**.

Fourth Embodiment

FIG. **10** illustrates an optical fiber cable temperature measurement device according to another embodiment of the present disclosure.

The optical fiber cable temperature measurement device includes: a power device **10** provided with a plurality of power device modules **20**; a plurality of optical fiber cable measurement units **430** that are provided in the respective power device modules **20**; optical fiber cables **40** inserted and installed in the respective optical fiber cable measurement units **430**; and a control unit **60** that is connected to one end of the optical fiber cables **40** to measure temperatures.

In this embodiment, differences from the previous embodiment will be described.

The optical fiber cable measurement unit **430** includes a cable mount **431** and **441** and a fiber-optic cable (not shown). The cable mount **431** and **441** includes an upper cable mount **431** and a lower cable mount **441** which correspond to each other in size. For example, the upper cable mount **431** and the lower cable mount **441** are of the same width and length.

The upper cable mount **431** and the lower cable mount **441** are built in the form of a frame with an opening **433** and **443**. The upper cable mount **431** may be built in the form described in the first embodiment. That is, they may be built in the form of a rectangular frame with a plurality of horizontal bars **442a** and a plurality of vertical bars **442b**. The upper cable mount **431** may correspond in size to the lower cable mount **441**.

Stoppers **444** are provided on a bottom vertical bar **442b** of the lower cable mount **441**.

Positional marks **445** where a fiber-optic cable **40** is mounted are formed on the lower cable mount **441** or the upper cable mount **431**.

The fiber-optic cable **40** is fixed and installed to where the positional marks **445** are indicated. For example, an adhesive (e.g., a double-face tape or a sticker) may be applied or attached to where the positional marks **445** are present.

Although not shown, the positional marks **445** may be formed in the form of grooves, as in the above-described embodiment.

Fifth Embodiment

FIG. **11** illustrates an optical fiber cable measurement unit **30** according to a further embodiment of the present disclosure. The optical fiber cable measurement unit **30** is provided to measure the temperature between the power device module **20** and an adjacent power device module **20**. The optical fiber cable measurement unit **30** is disposed in such

a way as to have surface contact with one surface (e.g., upper surface) of the power device module **20** or to have a predetermined gap with it, to measure the surface temperature of the power device module **20** or the temperature in the space between battery modules **20**.

The optical fiber cable measurement unit **30** may include a pair of panel members **31** and **35**, a fiber-optic cable **40**, and a connector **50**.

The panel members **31** and **35** are provided. The panel members **31** and **35** are provided such that the fiber-optic cable **40** is attached or inserted and installed to them. The panel members **31** and **35** may be made of a board- or film-like thin member.

The panel members **31** and **35** may be provided as a single body where the fiber-optic cable **40** is installed.

Moreover, the panel members **31** and **35** may be arranged side by side as a pair. That is, the panel members **31** and **35** may include an upper panel member **31** and a lower panel member **35**.

The panel members **31** and **35** are preferably made of a high thermal conductive material. For example, they may be made of a Teflon sheet.

The panel members **31** and **35** are preferably made in the form of a thin plate. Accordingly, the load on each power device module **20** or the rack **11** may be minimized, and the occupied space may be minimized.

The panel members **31** and **35** are made thin enough to be inserted between adjacent battery modules **20**. For example, the panel members **31** and **35** made with a thickness of 3 mm or less. Since the gap between battery modules **20** is small (e.g., 10 mm), the panel members **31** and **35** are made thin so as to be disposed between them.

The fiber-optic cable **40** is provided. The fiber-optic cable **40** is inserted and installed between a pair of panel members **31** and **35**. The fiber-optic cable **40** is laid over the surfaces of the panel members **31** and **35**. The fiber-optic cable **40** is laid in such a way as to be wound or bent between the pair of panel members **31** and **35**. That is, the fiber-optic cable **40** is laid in various shapes such that it is inserted a predetermined length into the panel members **31** and **35**. The fiber-optic cable **40** may overlap when bent.

As such, the temperature around the power device module **20** may be measured for each surface through the fiber-optic cable **40** laid on the panel members **31** and **35**. As the fiber-optic cable **40** is a line, it may have a plurality of measurement points when laid on the surfaces of the panel members **31** and **35**.

Accordingly, the fiber-optic cable **40** is laid in a bent shape to ensure that a plurality of measurement points are provided within the limited area of an upper portion of the power device module **20**.

FIG. **12** illustrates an exploded perspective view of the optical fiber cable measurement unit **30**.

The panel members **31** and **35** may include an upper panel member **31** and a lower panel member **35**.

The upper panel member **31** is made of a thin plate. The upper panel member **31** may be made of a film material.

The lower panel member **35** is made of a thin plate. The lower panel member **35** may be made of a film material.

The lower panel member **35** has an insertion groove **36** formed in it where the fiber-optic cable **40** is inserted and installed. The insertion groove **36** may be formed in a zigzag manner. The insertion groove **36** is formed in a bent shape.

The insertion groove **36** includes a straight portion **36a** or a curved portion **36b**. Here, the radius of curvature of the curved portion **36b** is preferably 20 times as large as or larger than the diameter of the fiber-optic cable **40**, consid-

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er the material characteristics of the fiber-optic cable 40. In general, the radius of curvature that allows the fiber-optic cable 40 to keep working properly is known to be 20 times as large as or larger than the cross-section diameter of the fiber-optic cable 40.

The fiber-optic cable 40 is inserted and installed into the insertion groove 36 of the lower panel member 35. That is, the fiber-optic cable 40 is laid in the shape of the insertion groove 36. Since the fiber-optic cable 40 is laid in a bent shape on a single surface, it can cover a length required for measurement at a plurality of measurement points, within a given unit area of the power device module 20. The fiber-optic cable 40 is laid longitudinally on the power device module 20, and therefore this ensures that a plurality of temperature measurement points are provided.

The connector 50 is provided on both ends of the fiber-optic cable 40. As the connector 50 is provided on each panel member 31 and 35, the optical fiber cable measurement units 30 may be provided as separate units. Also, the connector 50 may be set as a reference point for temperature measurement (wavelength measurement). Accordingly, it may be set as a reference point of each optical fiber cable measurement unit 30.

A connecting cable 41 and 45 is provided to connect the control unit 60 and the connector 50 or two adjacent connectors 50. The optical fiber cable measurement units 30 may be interconnected by the connecting cable 45.

The control unit 60 may be configured as a DTS server. The control unit 60 is provided to provide wavelength measurements in the fiber-optic cable 40 and convert them into temperatures.

A proper operating temperature may be set on the control unit 60, and an indication or an alert may be issued if this temperature is out of a limited temperature range. For example, a display unit 61 may be provided on a portion of the server of the control unit 60.

FIG. 13 illustrates a power device module 20. One side (for example, top) of the power device module 20 may be divided into a plurality of sections a, b, and c. The sections may be segments that are defined to physically separate battery cells (not shown) within the power device module 20 by partitions or dividing walls. Alternatively, the sections may be defined by virtually dividing the top of the power device module 20 into a predetermined number of segments. Although FIG. 5 depicts three separate sections by way of example, the number of sections may be set properly as needed.

The fiber-optic cable 40 may have a plurality of measurement points at which the temperatures in the sections a, b, and c of the power device module 20 are measured. The layout of the fiber-optic cable 40 may be determined in such a way that the measurement points are properly disposed on the sections a, b, and c.

Sixth Embodiment

FIGS. 14 and 15 illustrate an optical fiber cable measurement unit 30A according to a further embodiment of the present disclosure.

What applies to the optical fiber cable measurement unit 30 of the previous embodiment equally applies to the optical fiber cable measurement unit 30A of this embodiment, except for some details to be described henceforth.

The optical fiber cable measurement unit 30A may include an upper panel member 31A and a lower panel member 35A and a mounting plate 70.

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Mounting plate installation parts 32 and 37 formed as grooves are respectively installed on the panel members 31A and 35A so that the mounting plate 70 is inserted and installed into them. The mounting plate installation parts 32 and 37 may correspond in position or number to the sections a, b, and c of the power device module 20. For example, the upper panel member 31A may have a mounting plate installation part 32a formed in the section a, a mounting plate installation part 32b formed in the section b, and a mounting plate installation part 32c formed in the section c. Also, a plurality of mounting plate installation parts 32 and 37 may be formed in each section a, b, and c. This may mean that each section is subdivided.

The mounting plate 70 may include a wiring groove 71 formed around the circumference. The fiber-optic cable 40 may be inserted and disposed along the wiring groove 71.

Since the mounting plate 70 is inserted and installed into the mounting plate installation parts 32 and 37, it may be disposed to correspond to the sections a, b, and c of the power device module 20.

Flanges 72 and 73 protrude from the top and bottom of the wiring groove 71. The thickness of the flanges 72 and 73 may be equal to the depth of the mounting plate installation parts 32 and 37. The flanges 72 and 73 of the mounting plate 70 may be installed on the mounting plate installation parts 32 and 37 of the panel members 31A and 35A, thereby exposing the wiring groove 71 of the mounting plate 70 between the panel members 31A and 35A.

FIG. 16 is a top view of the optical fiber cable measurement unit according to the embodiment of FIG. 14 from which the upper panel member 31A and the mounting plate 70 are removed.

The fiber-optic cable 40 is laid in such a way as to be wound along the mounting plates 70 (to have a curved portion or a turnaround portion) in order to cover enough length within a limited area. Measurement points 40a to 40f are set on the fiber-optic cable 40 in order to measure temperatures in each section. The measurement points 40a to 40f may not be physical points, but may be values that are obtained based on a distance that is measured from scattering waves by the control unit 60. An input terminal connector 50a may serve as a starting point of the optical fiber cable measurement unit 30A. The measurement points 40a to 40f may be set at predetermined distances from the starting point. The temperatures in each section are measured from the measurement points. FIG. 8 depicts an example in which temperatures are measured for six sections.

Seventh Embodiment

FIG. 17 illustrates another example of the lower panel member 35B. The upper panel member may be built in the same manner as the lower panel member 35B, and they may be symmetrical.

A mounting plate installation part 38 where the mounting plate 70 is installed may be formed in such a way that a plurality of circular grooves are connected. That is, both sides of the mounting plate installation part 38 may be formed in a wavelike pattern. The radius of curvature of the wavelike pattern may be equal to the radius of curvature of the mounting plate 70.

Since a plurality of installation grooves are formed on the mounting plate installation part 38, the mounting plate 70 may be disposed properly depending on the number of sections.

In each embodiment, the panel members 31 and 35 may be made of different materials. For example, the upper panel

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member **31** may be made of an insulating material, and the lower panel member **35** may be made of a non-insulating material. In this way, the directionality of temperature measurement may be set because heat transfer is blocked by an insulating member. That is, effective thermal control for the energy storage device may be performed since temperature measurement is done on surfaces where the temperature is highest.

According to each embodiment of the present disclosure, it is possible to facilitate temperature measurement for a power device and thermal control by an optical fiber cable measurement unit provided for each power device module.

Moreover, the optical fiber cable temperature measurement device may be applied for each power device module configured in multiple stages, by means of optical fiber cable measurement units.

In addition, since the fiber-optic cable is laid in a bent shape on the optical fiber cable measurement unit, linear temperature distribution may be switched to planar temperature distribution.

Furthermore, temperature measurement may be done by measuring a distance with respect to a connector.

Furthermore, the optical fiber cable temperature measurement device is provided in the form of optical fiber cable measurement units where the fiber-optic cable is mounted, and therefore may be easily assembled and installed in a power device.

Furthermore, the optical fiber cable measurement units make measurements on a per-sheet or per-plane basis, thereby enabling effective control of temperatures between battery modules.

Furthermore, the optical fiber cable measurement units allow for measurement using a fiber-optic cable that is inserted and laid between films (upper and lower panels) that are tightly held together, and therefore provides high measurement accuracy and enables conversion into measurement positions based on distance, thereby facilitating the arrangement and control of the measurement points.

Since the fiber-optic cable is laid on a film surface with a predetermined area, it is possible to secure a radius of curvature required for measurement.

Since the fiber-optic cable is installed in an installation groove or a mounting plate formed (placed) on the film surface, a standardized temperature measurement position distribution can be configured, thereby allowing for efficient temperature management.

Since the upper and lower panels may work in one direction by using an insulating/non-insulating film, the directionality of temperature measurement can be achieved.

Since temperature measurement is possible with respect to a connector provided for each optical fiber cable measurement unit. Thus, measurements may be made at the same points on the upper surfaces of battery modules even if the gap between the battery modules varies.

The embodiments described above are embodiments for implementing the present invention, and various modifications and variations may be made by those of ordinary skill in the art to which the present invention pertains without departing from the essential characteristics of the present invention. Therefore, the embodiments disclosed in the present invention are not intended to limit the technical spirit of the present invention, but to explain, and the scope of the technical spirit of the present invention is not limited by these embodiments. That is, the protection scope of the present invention should be interpreted by the claims below,

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and all technical ideas within the scope equivalent thereto should be interpreted as being included in the scope of the present invention.

The invention claimed is:

1. An optical fiber cable temperature measurement device comprising:

a power device provided with a plurality of power device modules;

a plurality of optical fiber cable measurement units that are provided in the respective power device modules; optical fiber cables inserted and installed in the respective optical fiber cable measurement units; and

a control unit that is connected to one end of the optical fiber cables to measure temperatures,

wherein each of the optical fiber cable measurement units comprise a cable mount where the optical fiber cable is inserted and mounted,

wherein the cable mount comprises a mount body built in the shape of a rectangular frame and a crossbar disposed across the mount body, and

wherein each of the optical fiber cable measurement units is disposed to contact with one surface of the power device module or to have a predetermined gap with the power device module so that each of the optical fiber cable measurement units measures the surface temperature of the power device module or the temperature in a space between power device modules.

2. The optical fiber cable temperature measurement device of claim 1, wherein the cable mount has a plurality of openings.

3. The optical fiber cable temperature measurement device of claim 1, wherein the mount body has a stopper that keeps the cable mount from entering when inserted into the power device module.

4. The optical fiber cable temperature measurement device of claim 1, wherein the cable mount comprises an upper cable mount built in the form of a frame and a lower cable mount built in the form of a frame corresponding to the upper cable mount.

5. The optical fiber cable temperature measurement device of claim 4, wherein the upper cable mount and the lower cable mount are held together in such a way as to swing relative to each other.

6. The optical fiber cable temperature measurement device of claim 4, wherein positional marks where the optical fiber cable is mounted are formed on the lower cable mount or the upper cable mount.

7. The optical fiber cable temperature measurement device of claim 1, wherein the mount body comprises a pair of horizontal bars disposed in parallel and a pair of vertical bars disposed in parallel.

8. The optical fiber cable temperature measurement device of claim 7, wherein the crossbar is disposed in the middle part of the pair of horizontal bars.

9. The optical fiber cable temperature measurement device of claim 7, wherein a plurality of cable fixing grooves are formed on a lower surface of the crossbar for fixing the optical fiber cable.

10. The optical fiber cable temperature measurement device of claim 9, wherein a side cross-section of the cable fixing grooves is shaped like a segment of a circle.

11. The optical fiber cable temperature measurement device of claim 9, wherein the cable fixing grooves are also formed on the vertical bars.

12. The optical fiber cable temperature measurement device of claim 11, wherein the cable fixing grooves formed

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on the crossbar are parallel to the horizontal bars, and the cable fixing grooves formed on the vertical bars are slanted from the horizontal bars.

13. The optical fiber cable temperature measurement device of claim 1, wherein each of the optical fiber cable measurement unit comprises panel members, and the optical fiber cable is attached or inserted and installed to the panel member.

14. The optical fiber cable temperature measurement device of claim 13, wherein the panel members comprise an upper panel member and a lower panel member, and the optical fiber cable is inserted and installed between the upper panel member and the lower panel member.

15. The optical fiber cable temperature measurement device of claim 14, further comprising mounting plate installation parts that are formed as grooves on the upper panel member and the lower panel member and the mounting plate inserted into the mounting plate installation parts.

16. The optical fiber cable temperature measurement device of claim 15, wherein the mounting plate has a wiring groove formed around a circumference in such a way the optical fiber cable is wound thereabout.

17. The optical fiber cable temperature measurement device of claim 14, wherein the upper panel member and the lower panel member are made of a thin plate or a film.

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18. The optical fiber cable temperature measurement device of claim 17, wherein the upper panel member or the lower panel member has an insertion groove formed therein where the optical fiber cable is inserted and installed,

wherein the insertion groove has a straight portion or a curved portion.

19. The optical fiber cable temperature measurement device of claim 18, wherein a radius of curvature of the curved portion is 20 times as large as or larger than a diameter of the optical fiber cable.

20. The optical fiber cable temperature measurement device of claim 14, further comprising a connector coupled to both ends of the optical fiber cable and exposed out of the upper panel member and the lower panel member.

21. The optical fiber cable temperature measurement device of claim 20, further comprising a connecting cable for connecting the control unit and the connector or connecting the power device modules.

22. The optical fiber cable temperature measurement device of claim 20, wherein each of the power device modules is divided into a plurality of sections, and measurement points are set on the optical fiber cable at predetermined distances from the connector.

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